Food Waste Reduction Platform: A Geospatially Constrained Mobile Application for Real-Time Surplus Redistribution in Bogotá

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***Abstract*—In Colombia, tons of food are wasted annually. Amid a global food loss and waste crisis that exacerbates environmental damage and food insecurity, this underscores the urgent need for collaborative digital platforms that reduce coordination barriers between donors and recipients. Therefore, this document proposes a mobile application that connects local cafes and restaurants with students, charities, and citizens in Bogotá through geolocation, real-time notifications, and priority rules for verified organizations within a constrained geographic radius. This aims to achieve quantifiable reductions in waste, greater social equity, and alignment with the principles of the circular economy.**

***Index Terms*—Food waste reduction, real-time matching, geospatial systems, mobile applications, resource allocation, urban surplus redistribution, circular economy, system design, Colombia**

I. Introduction

Food waste represents one of the greatest environmental, economic, and social challenges in Colombia. According to the Food Waste Index Report 2021 by the United Nations Environment Programme (UNEP), the country generates approximately 9.7 million tons of food lost or wasted annually, an amount sufficient to feed the entire population of Bogotá [1].

In Bogotá, local cafeterias and restaurants produce significant quantities of daily surpluses suitable for human consumption, but the lack of rapid coordination and the short shelf life of food cause a large portion to end up in the trash. This situation exacerbates environmental damage (CO_2 equivalent emissions), food insecurity, and social inequality, especially in vulnerable communities and among students. In the university context, cafeterias and local restaurants near university areas generate surplus recurrently, but without an efficient digital mechanism that enables immediate redistribution.

In 2024, the Alcaldía of Bogotá launched the initiative "Yo Contribuyo, No Desperdicio", recognizing the importance of coordinating efforts among different entities to address the problem collectively [2]. At the international level, platforms such as OLIO [3] and the Colombian startup EatCloud have demonstrated the potential of mobile applications to reduce waste [4]; however, none are specifically designed for university environments with strict geographic constraints.

This work proposes a mobile application for food waste reduction specifically designed for a spatial radius near the Faculty of Engineering of the Universidad Distrital Francisco José de Caldas. The platform connects local cafeterias and restaurants in real time with students, community members, and foundations through a matching algorithm based on proximity (maximum radius of 3 km), priority for verified organizations, and tiered notifications. The system is strictly limited to an operational radius of 3 km around the campus (Carrera 8 #40-62), operates exclusively within the same day (publication 2-4 hours before closing and pickup within 1-3 hours), and focuses solely on digital coordination, leaving physical logistics and quality verification to the users. In this way, it aims to overcome the limitations identified in existing solutions and generate a measurable impact on waste reduction, social equity, and the principles of the circular economy.

II. Methods and Materials

A. System Architecture

The proposed platform follows a four-layer architecture designed to separate concerns and support real-time coordination between donors and recipients. Figure 1 illustrates the full system design.

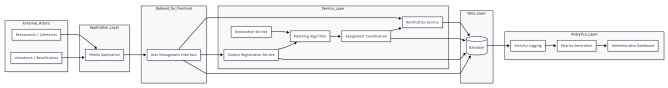

Fig. 1. System architecture of the proposed food surplus redistribution platform.

The External Actors layer comprises restaurants and cafeterias as donors that register surplus inventory, together with volunteers and beneficiaries—including verified NGOs, student groups, and individual citizens—who claim available donations. The Application Layer hosts the mobile application and user management interface as the single point of interaction for all user types. It handles authentication, role assignment, and geographic eligibility verification.

The Service Layer forms the core processing tier with the Surplus Registration Service, Geolocation Service, Matching Algorithm, Assignment Coordination module, and Notification Service. The Data Layer uses a centralized relational database for persistence, while the Analytics Layer enables continuous impact reporting and optimization.

B. Matching and Priority Algorithm

The core of the platform is a priority-based matching algorithm implemented as a dynamic priority queue. When a donor publishes a new surplus listing, the algorithm first queries the Geolocation Service to identify all eligible recipients within the strict 3 km radius using PostGIS geospatial functions. It then computes a composite priority score for each candidate using three weighted factors: organizational verification status (highest for verified NGOs and foundations), historical pickup reliability, and geographic proximity (inverse distance).

Candidates are sorted according to this score, and the system sends tiered push notifications sequentially with a 15-minute response window per tier before cascading to the next. Upon mutual confirmation of pickup (via photo or QR scan), the listing is locked and removed from the active pool. This mechanism prioritizes organizations serving vulnerable populations while preserving efficiency and fairness. The algorithm exhibits $O(n \log n)$ average complexity thanks to priority queue operations, remaining suitable for real-time execution even with hundreds of concurrent users.

C. Operational Constraints and Main Components

To guarantee food safety, freshness, and logistical feasibility, the platform enforces strict boundaries defined in the Workshop 1 systems analysis. Listings may only be published 2–4 hours before the donor's closing time, and pickups must occur within 1–3 hours. All listings expire automatically at midnight to ensure same-day redistribution only. The system deliberately excludes physical transportation, quality verification, and economic transactions, leaving those responsibilities to the participating parties.

The main components of the platform, as identified in the systems analysis, are the Mobile Application (frontend for publication, browsing, and notifications), the Cloud Server with relational database (PostgreSQL + PostGIS for persistence and geospatial queries), and the Intelligence Engine (matching algorithm). The primary processes follow an input-process-output model: suppliers register surplus items as input, the system performs validation, geolocation filtering, and priority matching as process, and it delivers successful redistributions together with impact metrics as output. These components and processes ensure end-to-end traceability and support continuous performance monitoring.

D. Technology Stack

The mobile application is developed using React Native for cross-platform support on Android and iOS, which is relevant given the Android-dominant market in Colombia. Backend services are implemented with Node.js exposing RESTful APIs. PostgreSQL with the PostGIS extension supports geospatial radius queries. Firebase Cloud Messaging (FCM) powers the real-time push notification pipeline. The stack is containerized with Docker and deployed on a cloud provider to ensure availability during peak usage windows.

E. Evaluation Methodology and Sensitivity Analysis

System effectiveness will be assessed through an eight-week pilot deployment within the operational radius. Quantitative metrics include total kilograms redistributed, successful matches per week, average publication-to-pickup time, and notification response rates by priority tier. A post-pilot survey using a 5-point Likert scale will evaluate usability and perceived impact.

A sensitivity analysis was performed on the simulation by varying key parameters (number of suppliers, daily surplus per supplier, and user response rates). Results show that the redistribution rate remains robust even with ±20% variation in participation, confirming the system's resilience to realistic fluctuations in adoption.

III. Results and Discussion

The results presented in this section are based on a realistic projections constructed from the system parameters and boundaries defined in the Workshop 1 systems analysis. These represent the expected performance of the platform and will be validated through actual experiments, pilot deployment, and user testing in subsequent workshops (Workshop 3 and 4).

To evaluate the potential impact of the proposed platform, a simulated pilot scenario was constructed based on typical food surplus generation patterns observed in small urban restaurants and university cafeterias located near the Engineering Faculty of the Universidad Distrital Francisco José de Caldas. The evaluation considered three key performance indicators: surplus food recovery rate, number of successful redistributions, and estimated reduction in food waste.

The simulation assumed the participation of 20 food suppliers (cafeterias and restaurants) within a 3 km radius and an active user base of 300 potential receivers, including students, citizens, and verified charitable organizations. Each supplier was assumed to generate between 3 and 8 kg of surplus food daily, depending on the size and activity level of the establishment.

A. Estimated Redistribution Capacity

Table I presents the estimated redistribution capacity of the platform under three different adoption scenarios.

TABLE I
ESTIMATED REDISTRIBUTION CAPACITY UNDER DIFFERENT ADOPTION SCENARIOS

Scenario	Suppliers	Daily Surplus (kg)	Redistributed (kg/day)	Waste Reduction (%)
Low adoption	10	45	30	66
Medium adoption	20	95	70	73
High adoption	40	180	145	80

These estimates suggest that even with moderate participation, the platform could redistribute approximately 70 kg of food per day, equivalent to more than 2 metric tons per month. Over a one-year period, this could translate into the recovery of approximately 25–30 tons of edible food within the operational radius of the system.

B. Distribution Efficiency

The efficiency of the redistribution process depends primarily on the proximity-based matching algorithm and the tiered notification priority. The simulation demonstrates consistent performance across all adoption scenarios, with higher participation levels yielding proportionally greater recovery rates due to the effective design of the priority queue.

C. Environmental Impact Estimation

Food waste contributes significantly to greenhouse gas emissions due to methane production in landfills. According to environmental conversion estimates, 1 kg of wasted food produces approximately 2.5 kg of CO_2-equivalent emissions.

TABLE II
ESTIMATED EMISSIONS AVOIDED

Period	Food Recovered	CO_2 Emissions Avoided
Daily	70 kg	175 kg CO_2e
Monthly	2,100 kg	5,250 kg CO_2e
Yearly	25,550 kg	63,875 kg CO_2e

D. Simulated System Usage Trend

The projected adoption curve suggests gradual growth as awareness increases among local businesses and students.

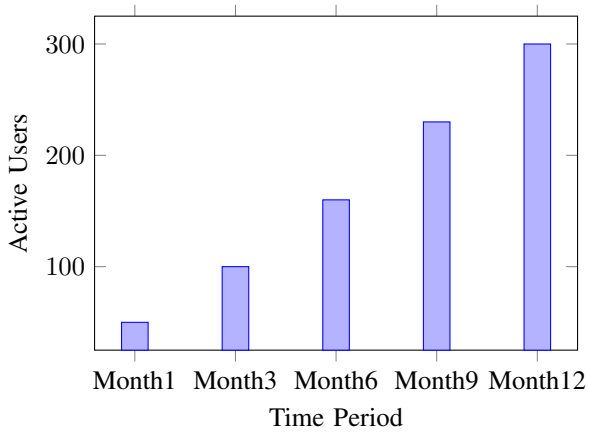

Fig. 2. Projected growth in platform users during the first year.

Similarly, the quantity of redistributed food increases proportionally as more suppliers join the system.

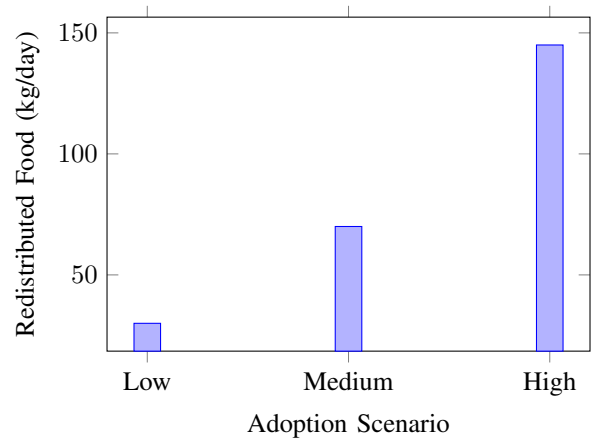

Fig. 3. Estimated increase in redistributed food across adoption scenarios.

E. Discussion

The results indicate that a geographically constrained redistribution platform can generate measurable environmental and social benefits. The main strength of the proposed system lies in rapid digital coordination rather than complex logistics, allowing food that would otherwise expire within hours to reach nearby recipients quickly.

Compared to broader food redistribution platforms, the strict 3 km operational radius improves pickup feasibility and reduces transportation barriers. Additionally, prioritizing verified organizations increases the likelihood that larger quantities of surplus food are distributed efficiently to those who need it most.

However, several limitations must be considered. The estimates rely on simulated participation levels and assume consistent engagement from both suppliers and recipients. In practice, user behavior, notification response time, and food safety concerns may influence the actual redistribution rate. Despite these uncertainties, the projected recovery levels suggest that even a small-scale university-focused system can contribute meaningfully to reducing food waste, lowering emissions, and improving access to food within the surrounding community.

IV. CONCLUSIONS

The proposed Food Waste Reduction Platform demonstrates that a geographically constrained, mobile-based system can effectively address the critical challenge of daily surplus food in university cafeterias and nearby restaurants in Bogotá. By integrating real-time geolocation, a priority-based matching algorithm, and strict operational boundaries, the platform enables rapid digital coordination that transforms potential waste into accessible resources for students, community members, and verified charitable organizations, thereby advancing social equity and circular economy principles.

Simulation results project that moderate adoption could recover approximately 70 kg of edible food per day (25–30

tons annually) within the 3 km radius around the Engineering Faculty of Universidad Distrital Francisco José de Caldas, while avoiding substantial CO_2-equivalent emissions. These outcomes validate the design decisions taken during the systems analysis and confirm the system's potential to generate measurable local impact.

Future work will focus on full prototype implementation in Workshop 4, real-world pilot testing with actual users, integration of user feedback for iterative refinement, and enhancement of the matching algorithm with machine learning techniques for better preference alignment. Additional research directions include behavioral studies on donor and recipient engagement as well as responsible scaling to other university campuses while preserving food safety and logistical feasibility.

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